Question Slides

LaTeXQuest

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Question 1 – Differential Equations

(i) You can go from Delhi to Jaipur either by car or by bus or by train or by air. In how many ways can you plan your journey from Delhi to Jaipur and back to Delhi?

(ii) Suppose you like variety and you don't want to return by the same mode of transport. How many different ways are possible?

[1.0 mark]

Solution 1

(i) There are 4 modes of transport available to go from Delhi to Jaipur and 4 modes available to return:

$$\text{Total ways} = 4 \times 4 = 16$$

(ii) There are 4 ways to go from Delhi to Jaipur. Since the return journey cannot use the same mode, there are $4 - 1 = 3$ modes left for returning:

$$\text{Total ways} = 4 \times 3 = 12$$

Question 2 – Calculus

(i) A lady wants to select one cotton saree and one polyester saree from a textile shop. If there are 10 cotton varieties and 12 polyester varieties, in how many ways can she choose the two sarees?

(ii) If in the above case, the lady has a limited budget, in how many ways can she choose one saree?

[1.0 mark]

Solution 2

(i) By the Fundamental Principle of Multiplication:

$$\text{Total ways} = 10 \times 12 = 120$$

(ii) She can choose either a cotton saree or a polyester saree. By the Fundamental Principle of Addition:

$$\text{Total ways} = 10 + 12 = 22$$

Question 3

In a class, there are 27 boys and 14 girls. In how many ways can the teacher form a team of one boy and one girl from amongst the students of the class to represent the class for a function?

[1.0 mark]

Solution 3

A boy can be selected in 27 ways and a girl can be selected in 14 ways:

$$\text{Total ways} = 27 \times 14 = 378$$

Question 4

Three persons enter a railway carriage, where there are 5 vacant seats. In how many ways can they seat themselves?

[1.0 mark]

Solution 4

The first person has 5 choices, the second person has 4 choices, and the third person has 3 choices:

$$\text{Total ways} = 5 \times 4 \times 3 = 60$$

Question 5

How many words can be formed out of the letters of the word **MAGIC** taking all the letters at a time (no letter being repeated)?

[1.0 mark]

Solution 5

The word **MAGIC** contains 5 distinct letters (M, A, G, I, C). The number of arrangements taking all 5 letters at a time is:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Question 6 – Trigonometry

How many 4-letter code words are possible using the first 10 letters of the English alphabet if

(i) no letter can be repeated?

(ii) letters may be repeated?

[1.0 mark]

Solution 6

(i) Without repetition:

$$\text{Total words} = 10 \times 9 \times 8 \times 7 = 5040$$

(ii) With repetition allowed:

$$\text{Total words} = 10 \times 10 \times 10 \times 10 = 10^4 = 10000$$

Question 7

How many automobile license plates can be made if each plate contains two different letters followed by three different digits?

[1.0 mark]

Solution 7

Assuming standard 26 English letters and 10 digits (0–9):

First letter: 26 ways, Second letter: 25 ways.

First digit: 10 ways, Second digit: 9 ways, Third digit: 8 ways.

$$\text{Total license plates} = (26 \times 25) \times (10 \times 9 \times 8) = 650 \times 720 = 468000$$

Question 8 – Trigonometry

How many 5-digit telephone numbers can be constructed using the digits 0 to 9 if each number starts with 67 and no digit appears more than once?

[1.0 mark]

Solution 8

The first two positions are fixed as 6 and 7 (1 way each).
Out of 10 digits (0–9), 2 digits are already used, leaving 8 available digits for the remaining 3 places:

$$\text{Total numbers} = 1 \times 1 \times 8 \times 7 \times 6 = 336$$

Question 9

In a certain city, all telephone numbers have six digits, the first two digits always being 41 or 42 or 46 or 62 or 64. How many telephone numbers have all six digits distinct?

[1.0 mark]

Solution 9

There are 5 possible choices for the two-digit prefix $(41, 42, 46, 62, 64)$. In each prefix, 2 distinct digits are used.
For the remaining 4 positions, digits must be chosen from the remaining $10 - 2 = 8$ distinct digits without repetition:

$$\text{Ways for remaining 4 places} = 8 \times 7 \times 6 \times 5 = 1680$$

$$\text{Total telephone numbers} = 5 \times 1680 = 8400$$

Question 10

How many 3-digit even numbers can be formed from the digits 1, 2, 3, 4, 5, 6 if the digits can be repeated?

[1.0 mark]

Solution 10

For a number to be even, the units place must be filled by 2, 4, or 6 (3 choices).
Since repetition is permitted:

Hundreds place: 6 choices

Tens place: 6 choices

Units place: 3 choices

$$\text{Total numbers} = 6 \times 6 \times 3 = 108$$

Question 11

How many numbers lying between 100 and 1000 can be formed with the digits 0, 1, 2, 3, 4, 5 if the repetition of digits is not allowed?

[1.0 mark]

Solution 11

Numbers lying strictly between 100 and 1000 are 3-digit numbers.

The hundreds place cannot be 0, so it can be filled by $\{1, 2, 3, 4, 5\}$ (5 choices).

The tens place can be filled by any of the remaining 5 digits including 0 (5 choices).

The units place can be filled by any of the remaining 4 digits (4 choices).

$$\text{Total numbers} = 5 \times 5 \times 4 = 100$$

Question 12

How many 4-digit numbers are there with no digit repeated?

[1.0 mark]

Solution 12

The available digits are $0, 1, 2, \ldots, 9$ (10 digits).

Thousands place cannot be 0: 9 choices (1–9).

Hundreds place: 9 choices (any remaining digit including 0).

Tens place: 8 choices.

Units place: 7 choices.

$$\text{Total numbers} = 9 \times 9 \times 8 \times 7 = 4536$$

Question 13

Find the number of 4-digit odd numbers, when repetition of digits is not allowed.

[1.0 mark]

Solution 13

The units place must be chosen from the odd digits $\{1, 3, 5, 7, 9\}$ (5 choices).
The thousands place cannot be 0 and cannot be the digit already chosen for the units place, leaving $10 - 2 = 8$ choices.
The hundreds place can be any remaining digit: $10 - 2 = 8$ choices.
The tens place can be any remaining digit: $10 - 3 = 7$ choices.

$$\text{Total numbers} = 5 \times 8 \times 8 \times 7 = 2240$$

Question 14 – Number Theory

How many numbers divisible by 5 and lying between 4000 and 5000 can be formed from the digits 2, 4, 5, 7 and 9, if

(i) repetition of digits is allowed?

(ii) repetition of digits is not allowed?

[1.0 mark]

Solution 14

A number between 4000 and 5000 has 4 digits with thousands digit equal to 4 (1 choice).
To be divisible by 5, the units place must be 0 or 5; from the given set $\{2, 4, 5, 7, 9\}$, the units digit must be 5 (1 choice).

(i) Repetition allowed:
Tens and hundreds places can each be filled by any of the 5 digits:

$$\text{Total numbers} = 1 \times 5 \times 5 \times 1 = 25$$

(ii) Repetition not allowed:
Digits 4 and 5 are fixed at the first and last positions respectively. The remaining 2 places must be filled from the remaining 3 digits $\{2, 7, 9\}$:

$$\text{Total numbers} = 1 \times 3 \times 2 \times 1 = 6$$

Question 15 – Number Theory

Find the number of positive integers greater than 6000 and less than 7000 which are divisible by 5, provided that no digit is to be repeated.

[1.0 mark]

Solution 15

The number must be a 4-digit integer starting with 6 (1 choice).
To be divisible by 5, the units place must be 0 or 5.

Case 1 (Units digit is 0):
Thousands place has 1 choice (6), units place has 1 choice (0).
The hundreds place can take any of the remaining $10 - 2 = 8$ digits, and the tens place can take any of the remaining 7 digits:

$$1 \times 8 \times 7 \times 1 = 56$$

Case 2 (Units digit is 5):
Thousands place has 1 choice (6), units place has 1 choice (5).
Similarly, hundreds place has 8 choices, tens place has 7 choices:

$$1 \times 8 \times 7 \times 1 = 56$$

$$\text{Total numbers} = 56 + 56 = 112$$

Question 16 – Number Theory

How many numbers divisible by 5 and lying between 40000 and 50000 can be formed from the digits 0, 3, 4, 5, 8 and 9, if

(i) repetition of digits is not allowed?

(ii) repetition of digits is allowed?

[1.0 mark]

Solution 16

The number is a 5-digit number strictly between 40000 and 50000, so the first digit must be 4. To be divisible by 5, the units digit must be 0 or 5.

(i) Repetition not allowed:

First digit: 1 choice (4).
Units digit: 2 choices (0 or 5).
Remaining 3 middle positions chosen from the remaining $6 - 2 = 4$ digits: $4 \times 3 \times 2 = 24$ ways.

$$\text{Total numbers} = 1 \times 4 \times 3 \times 2 \times 2 = 48$$

(ii) Repetition allowed:

First digit: 1 choice (4).
Middle 3 positions: 6 choices each.
Units digit: 2 choices (0 or 5).

This gives $1 \times 6 \times 6 \times 6 \times 2 = 432$ numbers in $[40000, 49995]$.
Since the question asks for numbers strictly *between* 40000 and 50000, the lower

Question 17

How many 5-digit even numbers can be formed from the digits 0, 1, 2, 3 and 4, if repetition of digits is not allowed?

[1.0 mark]

Solution 17

The available digits are $\{0, 1, 2, 3, 4\}$. An even number must end in 0, 2, or 4.

Case 1: Units digit is 0

The first 4 digits can be filled using the remaining 4 non-zero digits in:

$$4! = 24 \text{ ways}$$

Case 2: Units digit is 2 or 4 (2 choices)

The ten-thousands place cannot be 0 and cannot be the units digit: $5 - 2 = 3$ choices.

The remaining 3 positions can be filled by the remaining 3 digits in $3! = 6$ ways:

$$2 \times 3 \times 3! = 2 \times 3 \times 6 = 36 \text{ ways}$$

$$\text{Total numbers} = 24 + 36 = 60$$

Question 18

In how many ways can 5 persons sit in a car, 2 including the driver in the front seat and 3 in the back seat, if 2 particular persons do not know driving?

[1.0 mark]

Solution 18

The driver's seat must be occupied by one of the $5 - 2 = 3$ persons who can drive. After assigning the driver's seat, the remaining 4 persons can be seated in the remaining 4 seats (1 front, 3 back) in 4! ways:

$$\text{Total ways} = 3 \times 4! = 3 \times 24 = 72$$

Question 19

In how many ways can 2 prizes (in Science and Maths) be awarded to 15 students? In how many ways can the first and second prize in History be awarded to 15 students?

[1.0 mark]

Solution 19

Prizes in Science and Maths:
Assuming a student can receive both prizes:

$$15 \times 15 = 225 \text{ ways}$$

(If the same student cannot receive both prizes, it would be $15 \times 14 = 210$ ways).

First and second prize in History:
The same student cannot receive both first and second prize in the same subject:

$$15 \times 14 = 210 \text{ ways}$$

Question 20

Given 5 flags of different colours. How many different signals can be generated by hoisting the flags on a vertical pole (one below the other) if each signal requires the use of

(i) two flags?

(ii) at least two flags?

[1.0 mark]

Solution 20

(i) Signals using exactly 2 flags:

$$P(5,2) = 5 \times 4 = 20$$

(ii) Signals using at least two flags (i.e., 2, 3, 4, or 5 flags):

$$\text{Total} = P(5,2) + P(5,3) + P(5,4) + P(5,5) \quad = (5 \times 4) + (5 \times 4 \times 3) + (5 \times 4 \times 3$$

Question 21

How many numbers are there between 99 and 1000 such that

(i) every digit is either 2 or 5?

(ii) there is no restriction?

(iii) no digit is repeated?

(iv) the digit in the hundred's place is 5?

(v) the digit in the unit's place is 5?

(vi) at least one of the digits is 5?

(vii) at least one of the digits is repeated?

(viii) exactly one digit is 5?

[1.0 mark]

Solution 21

The numbers strictly between 99 and 1000 are the 3-digit numbers from 100 to 999.

(i) Each of the 3 positions has 2 choices ($\{2, 5\}$):

$$2 \times 2 \times 2 = 8$$

(ii) Hundreds digit $\in \{1, \ldots, 9\}$ (9 choices), tens digit (10 choices), units digit (10 choices):

$$9 \times 10 \times 10 = 900$$

(iii) First digit: 9 choices, second: 9 choices, third: 8 choices:

$$9 \times 9 \times 8 = 648$$

(iv) Hundreds place is fixed as 5 (1 choice):

$$1 \times 10 \times 10 = 100$$

Question 22

How many numbers are there between 100 and 1000 such that at least one of their digits is 3?

[1.0 mark]

Solution 22

Numbers strictly between 100 and 1000 are the 3-digit integers from 101 to 999 (total: $999 - 100 = 899$ numbers).

Numbers with no digit 3 in $[100, 999]$**:**
Hundreds place: 8 choices ($\{1, \ldots, 9\} \setminus \{3\}$).
Tens place: 9 choices ($\{0, \ldots, 9\} \setminus \{3\}$).
Units place: 9 choices ($\{0, \ldots, 9\} \setminus \{3\}$).
Total in $[100, 999] = 8 \times 9 \times 9 = 648$.

Since 100 contains no digit 3, the number of integers in the open interval $(100, 1000)$ with no 3 is $648 - 1 = 647$.

$$\text{Total with at least one } 3 = 899 - 647 = 252$$

Question 23

How many numbers can be formed from the digits 1, 2, 3, 9 if the repetition of digits is not allowed?

[1.0 mark]

Solution 23

Numbers can be 1-digit, 2-digit, 3-digit, or 4-digit:

1-digit numbers: $P(4,1) = 4$

2-digit numbers: $P(4,2) = 4 \times 3 = 12$

3-digit numbers: $P(4,3) = 4 \times 3 \times 2 = 24$

4-digit numbers: $P(4,4) = 4! = 24$

$$\text{Total numbers} = 4 + 12 + 24 + 24 = 64$$

Question 24

How many of the natural numbers from 1 to 1000 have none of their digits repeated?

[1.0 mark]

Solution 24

Since 1000 has repeated digits (0 repeats), we check 1-digit, 2-digit, and 3-digit numbers:

1-digit numbers (1–9): 9

2-digit numbers (10–99): $9 \times 9 = 81$

3-digit numbers (100–999): $9 \times 9 \times 8 = 648$

$$\text{Total numbers} = 9 + 81 + 648 = 738$$

Question 25 – Trigonometry

Find the total of natural numbers that can be formed by using the digits 0, 1, 2, 3, 4 and 5, if repetition of digits is not allowed.

[1.0 mark]

Solution 25

Natural numbers cannot have 0 as the leading digit.

1-digit numbers: $\{1, 2, 3, 4, 5\} \implies 5$

2-digit numbers: $5 \times 5 = 25$

3-digit numbers: $5 \times 5 \times 4 = 100$

4-digit numbers: $5 \times 5 \times 4 \times 3 = 300$

5-digit numbers: $5 \times 5 \times 4 \times 3 \times 2 = 600$

6-digit numbers: $5 \times 5 \times 4 \times 3 \times 2 \times 1 = 600$

$$\text{Total numbers} = 5 + 25 + 100 + 300 + 600 + 600 = 1630$$

Question 26 – Trigonometry

How many odd numbers less than 1000 can be formed using the digits 0, 1, 4, 5, 7, 8 if the repetition of digits is allowed?

[1.0 mark]

Solution 26

The available digits are $\{0, 1, 4, 5, 7, 8\}$ (6 digits).
Odd digits available are $\{1, 5, 7\}$ (3 choices for the units place).

1-digit odd numbers: 3 (namely 1, 5, 7)

2-digit odd numbers: Tens place cannot be 0 (5 choices), units place has 3 choices:

$$5 \times 3 = 15$$

3-digit odd numbers: Hundreds place cannot be 0 (5 choices), tens place has 6 choices, units place has 3 choices:

$$5 \times 6 \times 3 = 90$$

$$\text{Total odd numbers} = 3 + 15 + 90 = 108$$

Question 27 – Trigonometry

What is the sum of the digits in the units place of all the numbers formed using the digits 3, 4, 5, 6 taken all at a time?

[1.0 mark]

Solution 27

The total 4-digit numbers formed without repetition is $4! = 24$.
By symmetry, each of the 4 digits appears in the units place equally often:

$$\frac{24}{4} = 6 \text{ times}$$

The sum of the digits in the units place is:

$$6 \times (3 + 4 + 5 + 6) = 6 \times 18 = 108$$

Question 28

In how many ways can 5 different balls be distributed among 3 boxes?

[1.0 mark]

Solution 28

Each of the 5 distinct balls can be placed into any of the 3 boxes independently:

$$\text{Total ways} = 3^5 = 243$$

Question 29

Find the total number of ways of answering 6 multiple-choice questions, each question having 4 choices.

[1.0 mark]

Solution 29

Each of the 6 questions can be answered in 4 different ways:

$$\text{Total ways} = 4^6 = 4096$$

Question 30

Find the number of six-digit numbers, all digits of which are odd.

[1.0 mark]

Solution 30

The odd digits are $\{1, 3, 5, 7, 9\}$ (5 choices).
Each of the 6 digit positions must be an odd digit:

$$\text{Total numbers} = 5^6 = 15625$$

Question 31

In how many ways can 4 different balls be distributed in 5 boxes so that all the balls are not put in the same box?

[1.0 mark]

Solution 31

Each of the 4 balls has 5 choices of boxes, giving a total of:

$$5^4 = 625 \text{ distributions}$$

There are 5 ways where all 4 balls are put into the same box (all in box 1, all in box 2, ..., all in box 5).
Subtracting these cases:

$$\text{Total ways} = 5^4 - 5 = 625 - 5 = 620$$